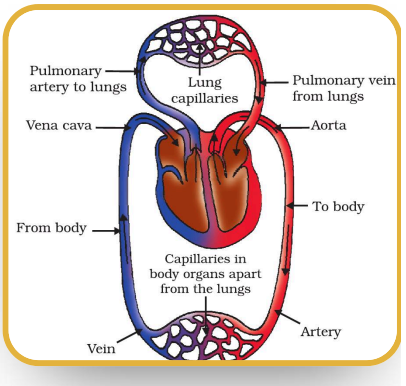
MT Classes, Motihari

Life Processes

Class 10 Science, Chapter 5 (NCERT, CBSE 2026-27)



Nutrition, respiration, transportation and excretion in plants and humans, explained beside every diagram of the textbook, with a photosynthesis lab, a food-journey tracker, a heartbeat route-finder, a nephron simulator, real CBSE board questions with answers, every NCERT question solved and a quiz.

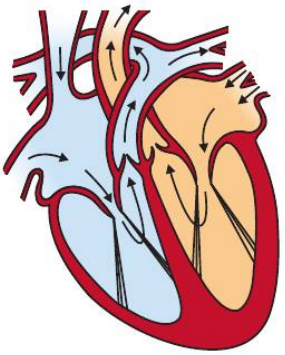
📌 Chapter snapshot

- Chapter 5 of NCERT Class 10 Science (reprint 2026-27), the **longest biology chapter** (21 pages). It is in Unit II "World of Living", which carries **25 marks** in the 80-mark theory paper with Control and Coordination, How do Organisms Reproduce? and Heredity.
- Asked **every year**: MCQs on **enzymes** (salivary amylase, trypsin, lipase), **pathways of respiration**, **stomata**; 2–3 mark questions on **digestion**, **alveoli**, **double circulation**, **xylem and phloem**, **nephron**; and case studies on **photosynthesis** (2025) and the **kidney** (2024).
- Diagrams to practise: **alimentary canal** (Fig. 5.6), **respiratory system** (5.9), **heart** (5.10), **nephron** (5.14), **excretory system** (5.13), **stomata** (5.3), **leaf section** (5.1).

5.1 What are life processes?

How do we know something is alive? We usually look for **movement**: breathing, growing, walking. But a plant that is not visibly growing is still alive, so visible movement alone is **not enough**.

- Biologists say living things always show **molecular movement** (molecules moving inside cells), even when asleep.
- **Viruses** show **no molecular movement** until they infect a cell. That is partly why there is a **controversy** about whether they are truly alive.
- Living bodies are highly **organised** (cells → tissues → organs). The environment keeps breaking this order down, so organisms must keep **repairing and maintaining** their structures, all the time.



Chapter opener: the heart, the pump that keeps every life process supplied

The processes which together perform this **maintenance job** are called **life processes**. They must go on even when we are just sitting in class or asleep.

🍎 Nutrition

Taking in food (a source of **energy** and **raw material**) from outside. Food is mostly **carbon-based**.

🫁 Respiration

Taking in **oxygen** and using it to break down food (oxidation–reduction reactions) to release **energy** for the cell.

❤️ Transportation

Carrying food and oxygen from the place they are taken in to **every cell**, and wastes away.

🚽 Excretion

Removing the harmful **by-products** of chemical reactions from the body.

🌱 Growth needs

If the body is to **grow**, extra raw material is also needed from outside.

⚡ Energy

All maintenance needs **energy**, which comes from **outside** the organism (food).

Why do big organisms need special organs?

- In a **single-celled** organism the whole surface touches the environment, so food intake, gas exchange and waste removal happen over the **entire surface** by diffusion.
- In **multicellular** organisms, most cells are deep inside, **not in direct contact** with the surroundings. **Simple diffusion will not meet the needs** of all the cells.
- So specialised tissues take in food and oxygen at one place, and a **transport system** carries them everywhere. Wastes need a specialised **excretory tissue** too.

> ⭐ **Exam line**
>
> Diffusion is insufficient in multicellular organisms like humans because the body is large, most cells are not in contact with the air, and diffusion over long distances is far too slow (oxygen would take about **3 years** to diffuse from lungs to toes, says NCERT). So we need lungs, blood, haemoglobin and a heart.



Nutrition: autotrophic nutrition

We need energy even when we do nothing, just to maintain order in the body. We also need materials to grow, develop and make proteins. Food supplies both. Organisms get food in two broad ways:

Autotrophs (auto = self)	Heterotrophs (hetero = other)
Use simple inorganic food CO_2 and water, using sunlight	**Use complex substances** broken down by bio-catalysts called **enzymes**
Examples Green plants, some bacteria	**Examples** Animals, fungi
Mode Photosynthesis	**Depend on** autotrophs, directly or indirectly

Photosynthesis

Photosynthesis is the process by which autotrophs take in carbon dioxide and water and convert them into **carbohydrates** in the presence of **sunlight** and **chlorophyll**.

$6CO_2 + 12H_2O$ —(chlorophyll, sunlight)→ $C_6H_{12}O_6 + 6O_2 + 6H_2O$

Carbon dioxide + water → glucose + oxygen + water. CBSE 2025 asked for exactly this equation.

- Carbohydrates are used for **energy**. Those not used immediately are stored as **starch**, the plant's internal energy reserve.
- Similarly, animals store some energy as **glycogen**. (CBSE 2023: plants store starch, animals glycogen.)

Three events of photosynthesis (learn in order)

1. **Absorption of light energy** by chlorophyll.
2. **Conversion of light energy to chemical energy** and **splitting of water** molecules into hydrogen and oxygen.
3. **Reduction of carbon dioxide** to carbohydrates.

🌵 Desert plants are different

These steps need not happen one after another immediately. **Desert plants take up CO_2 at night** (when it is cool, so opening stomata loses less water) and make an **intermediate**. During the day, the energy absorbed by chlorophyll acts on this intermediate. (Asked in CBSE 2023 and 2025.)

Where does it happen? The leaf (Fig. 5.1)

- In a cross-section of a leaf, some cells contain **green dots**: **chloroplasts**, which contain **chlorophyll**.
- Look for: waxy cuticle, upper epidermis, cells packed with chloroplasts, **air spaces**, lower epidermis with **guard cells** around stomata, and the vascular bundle (xylem and phloem) in the veins.

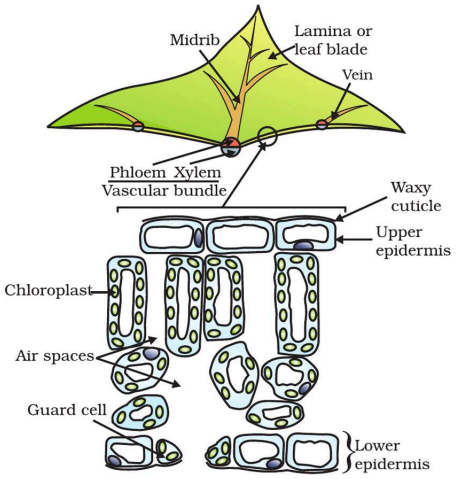


Fig. 5.1 Cross-section of a leaf

Is chlorophyll essential? Activity 5.1

1. Take a plant with **variegated leaves** (green and white patches), e.g. money plant or croton.
2. Keep it in a **dark room for 3 days** so all the starch is used up (destarching).
3. Keep it in **sunlight for about 6 hours**. Pluck a leaf and trace its green areas.
4. Dip the leaf in **boiling water** (kills cells, softens it), then **boil in alcohol on a water-bath**: the leaf turns pale and the alcohol turns **green** (chlorophyll dissolves out).
5. Dip it in **dilute iodine**, rinse and compare with the tracing.

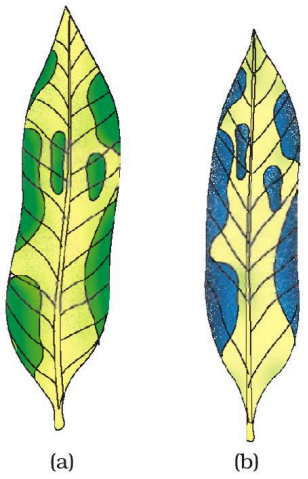


Fig. 5.2 Variegated leaf (a) before and (b) after starch test

⭐ Result and conclusion

Only the parts that were **green** turn **blue-black** with iodine (starch present). The white parts stay **brown** (no starch). So **chlorophyll is essential for photosynthesis**. Alcohol is heated on a **water-bath**, never directly, because it catches fire easily.

How does the plant get CO_2? Stomata (Fig. 5.3)

- **Stomata** are tiny pores on the surface of leaves. **Massive gas exchange** for photosynthesis takes place through them. (Gas exchange also happens across the surface of stems, roots and leaves.)
- Large amounts of **water are lost** through stomata too, so the plant **closes** them when it does not need CO_2.
- Opening and closing is done by the two **guard cells**: when **water**

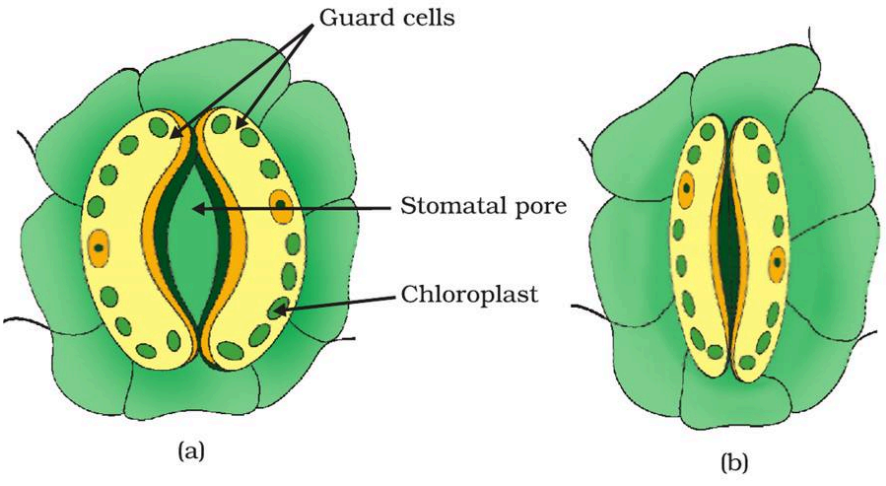


Fig. 5.3 (a) Open and (b) closed stomatal pore

flows into them they **swell** and the pore **opens**; when they **shrink**, the pore **closes**.

Is CO_2 essential? Activity 5.2

- Two similar potted plants, **destarched** (3 days in the dark).
- Each is placed on a glass plate under a **bell-jar**, sealed with **vaseline** to make it air-tight.
- Jar (a) has a watch-glass of **potassium hydroxide (KOH)**, which **absorbs CO_2**. Jar (b) has no KOH.
- After about 2 hours in sunlight, test a leaf from each for starch.

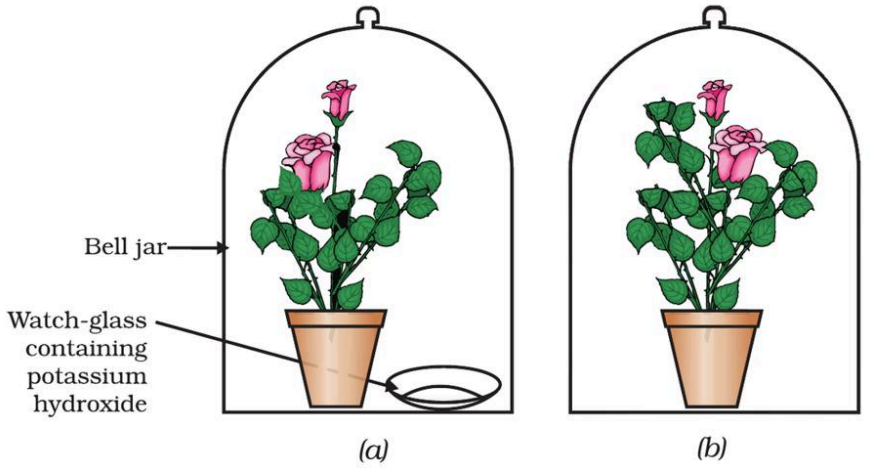


Fig. 5.4 Experimental set-up (a) with potassium hydroxide (b) without potassium hydroxide

⭐ **Result and conclusion**

The leaf from jar (a) shows **no starch** (stays brown): KOH removed the CO_2. The leaf from jar (b) turns **blue-black**. So **carbon dioxide is essential for photosynthesis**.

Other raw materials

- **Water** is taken up from the soil by the **roots**.
- **Nitrogen, phosphorus, iron, magnesium** are also taken from the soil. **Nitrogen** is needed to make **proteins**; it is taken as inorganic **nitrates or nitrites**, or as organic compounds made by **bacteria** from atmospheric nitrogen.
- CO_2 comes from the **air** through stomata; sunlight from the Sun; chlorophyll is in the chloroplasts.

5.2.2 Heterotrophic nutrition

Each organism is adapted to its environment. Food may be **stationary** (grass for a cow) or **mobile** (a deer for a lion), so the nutritive apparatus differs. There is a range of strategies:

Strategy	How food is taken	Examples (NCERT)
Saprophytic	Break down food **outside the body**, then absorb it	Fungi: bread mould, yeast, mushrooms
Holozoic	Take in **whole material** and break it down **inside** the body	Amoeba, humans, most animals
Parasitic	Get nutrition from plants or animals (hosts) **without killing them**	Cuscuta (amar-bel), ticks, lice, leeches, tapeworms

The names "saprophytic" and "holozoic" are not printed in this NCERT chapter but are widely used; NCERT itself names "parasitic".

5.2.3 How do organisms obtain their nutrition?

- In single-celled organisms, food may be taken in by the **entire surface**.
- **Amoeba** (Fig. 5.5): (a) senses a food particle; (b) puts out temporary finger-like extensions of the cell surface, **pseudopodia**; (c) these **fuse over** the food, forming a **food vacuole**; complex substances are broken down inside it and **diffuse into the cytoplasm**; (d) the undigested material moves to the surface and is **thrown out**.
- **Paramoecium** has a definite shape, and food is taken in at a **specific spot**. Food is moved to this spot by **cilia**, which cover the whole cell surface.

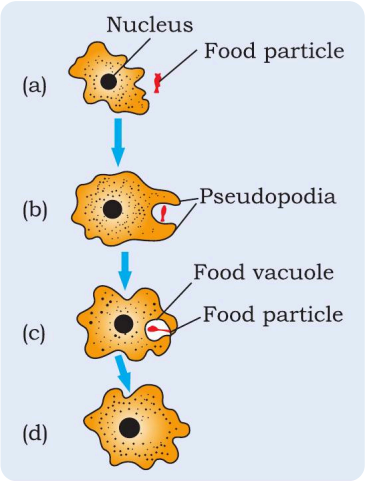


Fig. 5.5 Nutrition in Amoeba

🧠 **Memory trick for Amoeba: "I D A A E"**

Ingestion (pseudopodia engulf) → **D**igestion (in food vacuole) → **A**bsorption (into cytoplasm) → **A**ssimilation (used by the cell) → **E**gestion (waste thrown out). CBSE 2024 asked how Amoeba takes in and digests food.

5.2.4 Nutrition in human beings

The **alimentary canal** is a long tube from the **mouth to the anus**. Each region is specialised for a different job.

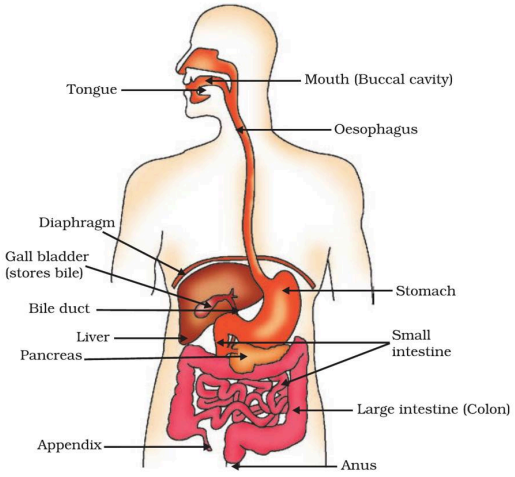


Fig. 5.6 Human alimentary canal. Learn all 14 labels: this is a favourite 5-mark diagram.

Mouth: teeth, saliva and tongue

- Food is **crushed by teeth** into small particles of the same texture.
- Food is wetted by **saliva** from the **salivary glands** to make its passage smooth. (Mouth "waters" when we see food we like.)
- Saliva contains the enzyme **salivary amylase**, which breaks down **starch** (complex) into **simple sugar**.
- The muscular **tongue** mixes food with saliva while chewing.

– **Activity 5.3: Starch solution in tubes A (with saliva) and B (without). After 20–30 min, add iodine. What happens?**

Tube B turns **blue-black** (starch still present). **Tube A** shows **no blue-black colour** (stays brownish-yellow): saliva has digested the starch. So **salivary amylase in saliva breaks down starch into sugar**.

Peristalsis and the oesophagus

- The lining of the canal has muscles that **contract rhythmically** to push food forward: **peristaltic movements**, all along the gut.
- Food goes from the mouth to the stomach through the **food-pipe (oesophagus)**.

Stomach

- A large organ that **expands** when food enters. Its muscular walls mix food with digestive juices.
- **Gastric glands** in its wall release three things:
  - **Hydrochloric acid**: creates an **acidic medium** that helps the enzyme pepsin act (it also kills germs in the food).

    - **Pepsin**: a **protein-digesting** enzyme.
    - **Mucus**: **protects the inner lining** of the stomach from the acid.

- Too much acid causes "**acidity**".
- Exit of food from the stomach is controlled by a **sphincter muscle**, which releases it in small amounts into the small intestine.

Small intestine: where digestion is completed

- The **longest part** of the canal, fitted into a small space by extensive **coiling**.
- Its length depends on diet: **herbivores** eating grass need a **longer** small intestine to digest **cellulose**; **carnivores** like tigers have a **shorter** one, as meat is easier to digest.
- It is the site of **complete digestion** of carbohydrates, proteins and fats. It receives secretions of the **liver** and **pancreas**.
- **Bile juice** (from the liver, stored in the gall bladder) makes the acidic food from the stomach **alkaline** so pancreatic enzymes can act. **Bile salts** break large fat globules into smaller ones (**emulsification**), like soap on dirt, which helps enzymes act.
- **Pancreatic juice** contains **trypsin** (digests **proteins**) and **lipase** (breaks down **emulsified fats**). (CBSE 2026 MCQ.)
- **Intestinal juice** from glands in the wall finally converts proteins → **amino acids**, complex carbohydrates → **glucose**, fats → **fatty acids and glycerol**. (CBSE 2024 MCQ.)
- Digested food is absorbed by the walls. The inner lining has numerous finger-like **villi**, which **increase the surface area** for absorption and are richly supplied with **blood vessels** that carry food to every cell.

Large intestine

- Unabsorbed food enters the large intestine, where its walls **absorb more water**.
- The rest is removed through the **anus**; its exit is regulated by the **anal sphincter**.

Digestive juices at a glance

Gland / organ	Juice	Contains	Action
Salivary glands	Saliva	Salivary amylase	Starch → simple sugar
Gastric glands (stomach)	Gastric juice	HCl, pepsin, mucus	Acid medium; proteins digested by pepsin; mucus protects lining
Liver	Bile	Bile salts (no enzyme)	Makes food alkaline; emulsifies fats

Gland / organ	Juice	Contains	Action
Pancreas	Pancreatic juice	Trypsin, lipase	Proteins digested; emulsified fats broken down
Intestinal glands	Intestinal juice	Enzymes	Final conversion to amino acids, glucose, fatty acids and glycerol

The pancreas also makes amylase for starch; this is extra information not in the NCERT text.

🦷 More to know: dental caries

Tooth decay is the gradual softening of **enamel and dentine**. Bacteria acting on **sugars** produce **acids** that demineralise enamel. Masses of bacteria and food particles stick to teeth as **dental plaque**; saliva cannot reach the tooth surface to neutralise the acid. **Brushing after eating** removes plaque before acids are made. If untreated, microbes may invade the **pulp**, causing inflammation and infection.

5.3 Respiration

– **Activity 5.4: Blow exhaled air into lime water, and pump ordinary air into lime water with a pichkari. Which turns milky faster?**

Lime water turns **milky much faster** with **exhaled air**. Lime water turns milky with CO_2 (calcium carbonate forms). So the air we breathe out contains **much more carbon dioxide** than ordinary air.

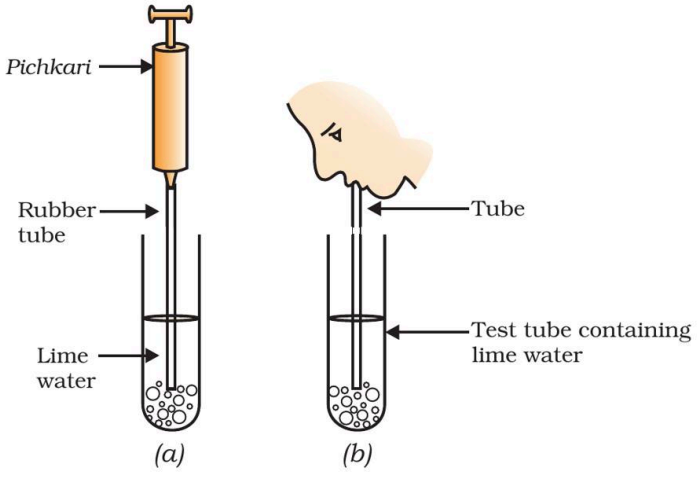


Fig. 5.7 (a) Air passed into lime water with a pichkari/syringe, (b) exhaled air passed into lime water

– **Activity 5.5: Yeast in sugar solution, with the gas led into lime water. What happens?**

Lime water turns **milky** after some time (usually within an hour, depending on temperature). So **fermentation by yeast produces carbon dioxide** (along with ethanol).

Food is used in cells to release energy. This is **cellular respiration**. In all organisms the **first step** is the same:

Glucose (6-carbon) —(in cytoplasm)→ Pyruvate (3-carbon) + energy

What happens to pyruvate next depends on oxygen:

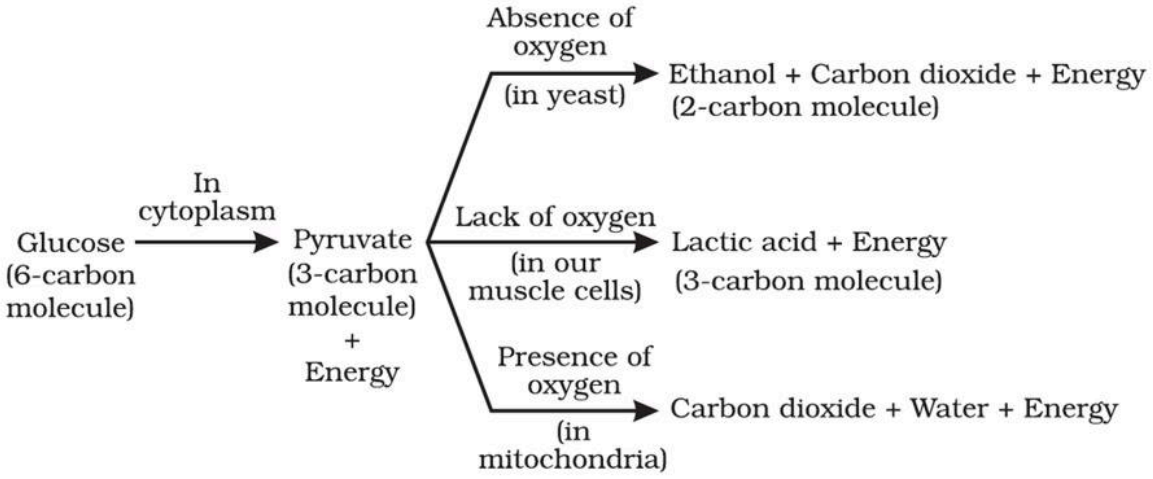


Fig. 5.8 Break-down of glucose by various pathways

🍺 **Anaerobic (yeast)**

No oxygen. Pyruvate → **ethanol** + CO_2 + energy. Called **fermentation**. Ethanol is 2-carbon.

🏃 **Lack of oxygen (muscles)**

During sudden heavy exercise. Pyruvate → **lactic acid** (3-carbon) + energy. Build-up of lactic acid causes **cramps**.

🔋 **Aerobic (mitochondria)**

Oxygen present. Pyruvate → **3** CO_2 + **water** + energy, in **mitochondria**. Releases **much more** energy.

🔋 More to know: ATP, the energy currency

Energy released in respiration is used immediately to make **ATP** (adenosine triphosphate) from **ADP + inorganic phosphate (Pi)**:

ADP + Pi —(energy)→ ATP

When the terminal phosphate bond of ATP is broken using water, **30.5 kJ/mol** of energy is released. Like a battery, ATP is used for **muscle contraction, protein synthesis, conduction of nerve impulses** and much more. It drives the **endothermic** reactions of the cell.

Gas exchange in plants

- Plants exchange gases through **stomata**, and large **inter-cellular spaces** keep all cells in contact with air. CO_2 and O_2 move by **diffusion**.
- The direction depends on **environmental conditions** and the **requirements of the plant** (CBSE 2023).

Breathing in water and on land

- Aquatic animals use **oxygen dissolved in water**. Dissolved oxygen is **fairly low** compared with air, so their **breathing rate is much faster** than that of land animals.
- **Fishes** take in water through the **mouth** and force it past the **gills**, where blood takes up the dissolved oxygen.
- Land animals breathe oxygen from the atmosphere. All breathing organs have a **large surface area**, a **very thin, delicate** surface, are placed **inside the body** for protection, and have **passages** and a **mechanism** to move air in and out.

5.3 The human respiratory system

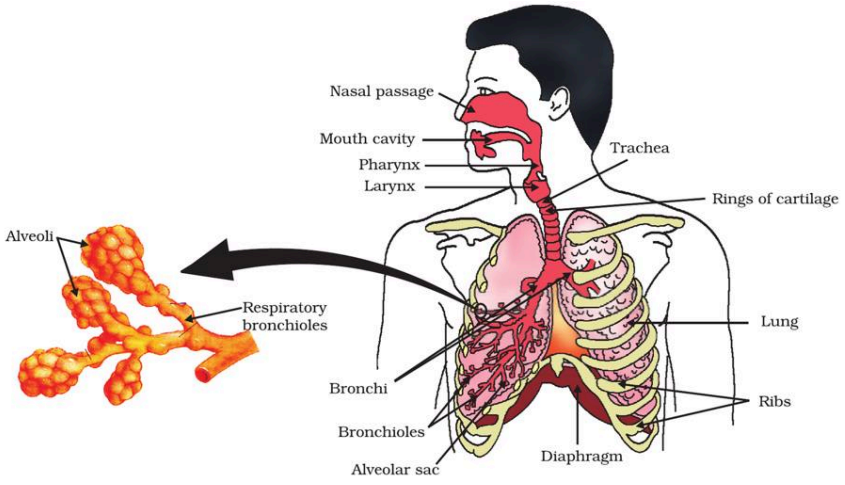


Fig. 5.9 Human respiratory system

The path of air (learn this flow)

Nostrils: air filtered by fine hairs and mucus

↓

Pharynx (throat) and larynx

↓

Trachea: rings of cartilage stop the air passage from collapsing

↓

Bronchi → bronchioles (smaller and smaller tubes)

↓

Alveoli: balloon-like sacs where gases are exchanged

- The **walls of the alveoli** have an extensive network of **blood vessels**.
- **Breathing in:** we **lift our ribs** and **flatten the diaphragm**, so the chest cavity becomes **larger**; air is sucked into the lungs and fills the expanded alveoli.
- Blood brings **CO_2** from the body for release into the alveoli; **oxygen** in the alveolar air is taken up by blood and carried to all cells.
- The lungs always contain a **residual volume** of air, so there is **enough time** for O_2 to be absorbed and CO_2 released.

Transport of gases

- In large animals, diffusion pressure alone cannot deliver oxygen to all parts. **Respiratory pigments** take up oxygen in the lungs and release it in oxygen-deficient tissues.
- In humans the pigment is **haemoglobin**, present in **red blood corpuscles**, with a **very high affinity for oxygen**. (CBSE 2026: not for CO_2!)
- **CO_2** is more soluble in water than oxygen, so it is mostly transported **dissolved in blood** (plasma).

📏 **80 m²**

Area of the alveolar surface if spread out, about 40 times the surface of your skin (which is roughly 2 m^2).

⏳ **3 years**

Time oxygen would take to diffuse from the lungs to the toes without haemoglobin and blood flow.

🚭 **Do you know? Smoking and tobacco**

The upper respiratory tract has hair-like **cilia** that remove germs, dust and particles from inhaled air. **Smoking destroys these cilia**, so germs, smoke and chemicals enter the lungs, causing infection, cough and even **lung cancer**. Tobacco in any form (cigarettes, bidis, hookah, **gutkha**) harms the tongue, lungs, heart and liver; India has a high rate of **oral cancer** from chewing gutkha. Just say NO to tobacco.



Transportation in human beings

Blood

- Blood is a **fluid connective tissue**: a fluid medium, **plasma**, with cells suspended in it.
- **Plasma** carries food, CO_2 and nitrogenous wastes in **dissolved** form, plus salts. **Red blood corpuscles** carry **oxygen**.
- So we need: a **pump** (heart), a **network of tubes** (blood vessels), and a **repair system** (platelets).

– **Activity 5.7: Find out the normal haemoglobin range in humans and in cattle. Is it different in men, women and children?**

Typical values from a health centre: adult men about **13–17 g/dL**, adult women about **12–15 g/dL**, children about **11–13 g/dL** (ranges differ slightly between labs). Men usually have more haemoglobin than women because of the male hormone testosterone, which increases red blood cell production, and because women lose blood every month in menstruation. In cattle and buffaloes the normal range is roughly **8–15 g/dL**, and differences between males, females and calves are explained in the same way (hormones, pregnancy, milk production and age). These are reference figures to compare with what you find; they are not given in NCERT.

Our pump: the heart (Fig. 5.10)

- A **muscular organ** about the size of our **fist**.
- Has **four chambers** so that **oxygen-rich blood does not mix** with CO_2-rich blood.
- **Atria** (upper chambers) are **thin-walled**: they only collect blood. **Ventricles** (lower) have **thicker muscular walls** because they pump blood to other organs.
- The **septum** separates the right and left sides.
- **Valves** stop blood flowing **backwards** when atria or ventricles contract.

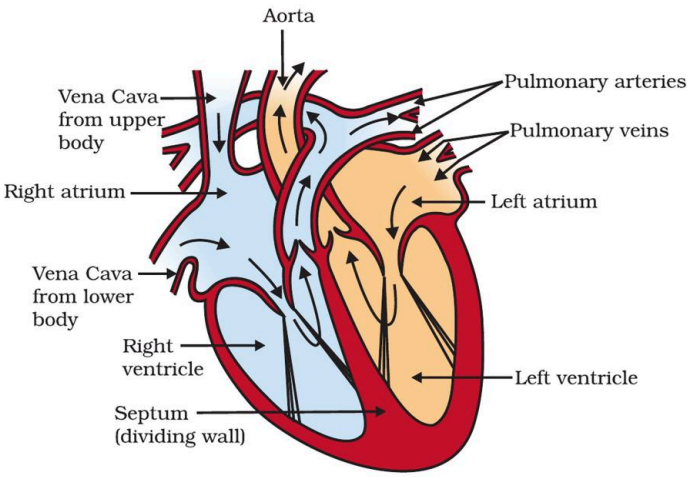


Fig. 5.10 Schematic sectional view of the human heart

🧠 Memory trick

Arteries carry blood **A**way from the heart. **V**eins bring blood back (**V** for "visit home"). Exception pair: the **pulmonary artery** carries **deoxygenated** blood (to the lungs) and the **pulmonary vein** carries **oxygenated** blood (from the lungs). CBSE 2023 asked about the vena cava and pulmonary artery.

Step by step: double circulation (Fig. 5.11)

1. Oxygen-rich blood from the **lungs** comes by the **pulmonary veins** to the **left atrium**, which relaxes while collecting it.
2. Left atrium **contracts**; the **left ventricle** relaxes and receives the blood.
3. The muscular left ventricle contracts and pumps blood through the **aorta** to the **body**.
4. Deoxygenated blood from the body comes by the **vena cava** to the **right atrium**.
5. Right atrium contracts; the **right ventricle** dilates and receives it.
6. Right ventricle pumps it through the **pulmonary artery** to the **lungs** for oxygenation.

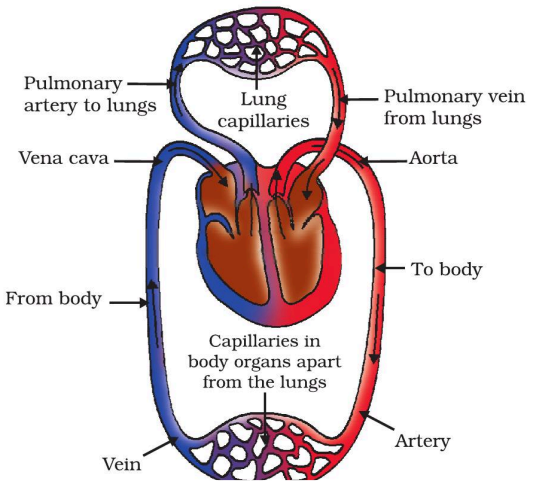


Fig. 5.11 Schematic representation of transport and exchange of oxygen and carbon dioxide

Blood passes through the heart **twice** in each complete cycle (heart → lungs → heart → body → heart). This is **double circulation**.

Why separate oxygenated and deoxygenated blood?

- Separation gives a **highly efficient supply of oxygen** to the body.
- This is useful in animals with **high energy needs**, like **birds and mammals**, which constantly use energy to **maintain body temperature** (warm-blooded).

Animal	Chambers	Mixing of blood?	Circulation
Fishes	2	Only deoxygenated blood passes through the heart	**Single**: heart → gills → body
Amphibians, many reptiles	3	Some mixing tolerated	Double (incomplete)
Birds, mammals	4	No mixing	Double (complete)

Amphibians and reptiles do not use energy to keep warm: their body temperature depends on the surroundings (cold-blooded).

🩺 More to know: blood pressure

- **Blood pressure** is the force that blood exerts against the wall of a vessel. It is **much greater in arteries** than in veins.
- **Systolic pressure**: in the artery during ventricular **contraction** (normal about **120 mm Hg**). **Diastolic pressure**: during ventricular **relaxation** (normal about **80 mm Hg**).
- Measured with a **sphygmomanometer**.
- High blood pressure is **hypertension**, caused by **constriction of arterioles**, which increases resistance to blood flow. It can lead to the **rupture of an artery** and internal bleeding.

The tubes: blood vessels

Arteries	Veins	Capillaries
Carry blood **away from** the heart to organs	Bring blood **back to** the heart	Connect arteries and veins inside organs
Thick, elastic walls (blood is under high pressure)	**Thin** walls (blood no longer under pressure)	Walls only **one cell thick**
No valves	Have **valves** so blood flows one way	Exchange of materials with cells happens here
Usually oxygenated blood (except pulmonary artery)	Usually deoxygenated (except pulmonary vein)	Join to form veins

🩹 Maintenance by platelets

If a vessel leaks, blood is lost and **pressure falls**, reducing the pumping efficiency. **Platelets** circulate and **plug leaks** by helping blood to **clot** at the injury. (CBSE 2025.)

💧 Lymph (tissue fluid)

Some plasma, proteins and blood cells escape through pores in capillaries into tissue spaces: this is **lymph**. It is like plasma but **colourless** with **less protein**. It drains into lymphatic capillaries → lymph vessels → large veins. Lymph **carries digested fat** from the intestine and **drains excess fluid** back into the blood.

5.4.2 Transportation in plants

- Leaves make food; roots take in water and minerals from the soil. If these are far apart (tall plants), **diffusion is not enough**, so a transport system is essential.
- Plants **do not move** and have many **dead cells**, so they have **low energy needs** and use **relatively slow** transport systems (CBSE 2025). But distances can be very large, as in tall trees.
- Two independent conducting tubes: **xylem** moves **water and minerals** from the soil; **phloem** moves **products of photosynthesis** from leaves to other parts.

Transport of water

1. In xylem, **vessels and tracheids** of roots, stems and leaves are interconnected into a continuous water channel.
2. Root cells in contact with soil **actively take up ions**. This creates a concentration difference between root and soil.
3. Water moves into the root to remove this difference: a steady movement into root xylem pushes a column of water **upwards** (**root pressure**).
4. Root pressure alone cannot lift water up tall trees. **Evaporation** of water from leaf cells creates a **suction** that pulls water up from the root xylem: **transpiration pull**.

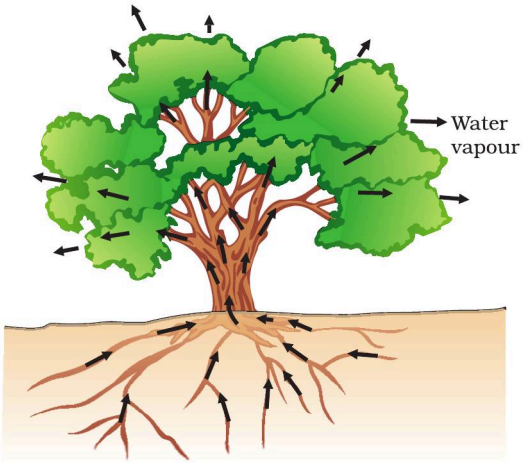


Fig. 5.12 Movement of water during transpiration in a tree

Transpiration is the loss of water as vapour from the aerial parts of a plant. It **helps absorption and upward movement** of water and dissolved minerals, and helps in **temperature regulation**.

⭐ Root pressure vs transpiration pull

Root pressure is more important at **night**. During the **day**, when stomata are open, **transpiration pull** is the **major driving force** for water in the xylem.

– **Activity 5.8: A potted plant and a stick, both covered with plastic sheets in sunlight for half an hour. Any difference?**

Water droplets appear on the inside of the plastic covering the **plant**, but not on the one covering the **stick**. The soil was covered, so the water came out of the **plant's leaves**: plants lose water vapour by **transpiration**.

Transport of food: translocation

- Transport of soluble products of photosynthesis is called **translocation**. It occurs in **phloem**.
- Phloem also carries **amino acids** and other substances, especially to **storage organs** (roots, fruits, seeds) and **growing organs**.
- It happens in **sieve tubes** with the help of adjacent **companion cells**, in **both upward and downward** directions.
- Unlike xylem, phloem transport **uses energy**: material like **sucrose** is loaded into phloem using **ATP**. This raises the **osmotic pressure**, water moves in, and the pressure pushes material to tissues with **less pressure**.
- So phloem moves material **according to the plant's needs**: in **spring**, sugar stored in root or stem is sent to the **buds**, which need energy to grow.

5.5 Excretion

Excretion is the biological process of removing harmful **metabolic wastes**, especially **nitrogenous** wastes, from the body. Many unicellular organisms simply let wastes **diffuse** from the body surface into the surrounding water.

5.5.1 Excretion in human beings

- Organs: a pair of **kidneys**, a pair of **ureters**, a **urinary bladder** and a **urethra**.
- Kidneys lie in the **abdomen**, one on either side of the **backbone**.
- Urine made in the kidneys → **ureters** → **urinary bladder** (stored) → released through the **urethra**.
- The purpose of making urine is to **filter out waste** such as **urea or uric acid** from the blood, just as CO_2 is removed in the lungs.

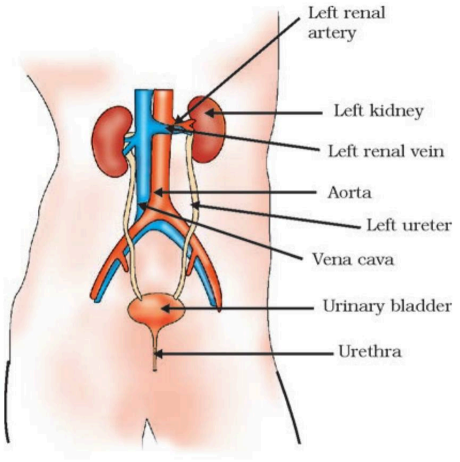


Fig. 5.13 Excretory system in human beings

The nephron: the filtration unit (Fig. 5.14)

- Each kidney has **large numbers** of filtration units called **nephrons**, packed close together.
- Like the lungs, the basic filter is a **cluster of very thin-walled capillaries**, the **glomerulus**.
- It sits in the cup-shaped end of a coiled tube, **Bowman's capsule**, which collects the **filtrate**.
- As filtrate flows along the **tubular part**, useful substances such as **glucose, amino acids, salts and a major amount of water** are **selectively re-absorbed** into the surrounding capillaries.
- Urine from each kidney enters a long tube, the **ureter**, to the bladder.

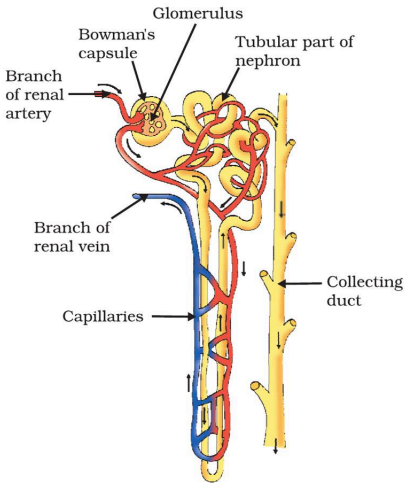


Fig. 5.14 Structure of a nephron

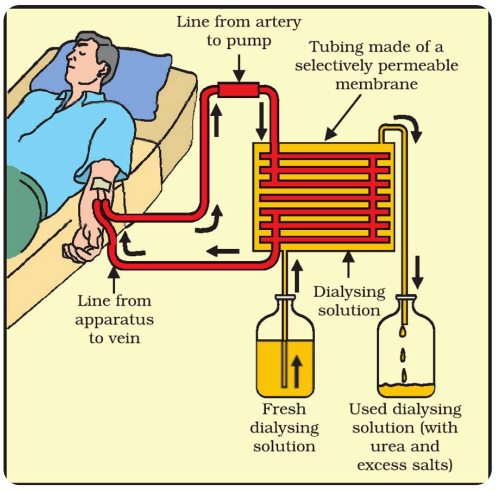
📌 How is the amount of urine regulated?

The amount of water re-absorbed depends on (1) **how much excess water** there is in the body, and (2) **how much dissolved waste** has to be excreted. Drink a lot of water → less is re-absorbed → more, dilute urine. Hot day, lots of sweating → more re-absorbed → less, concentrated urine. (CBSE 2024 case question.)

- The urinary bladder is **muscular** and under **nervous control**, so we can usually control the urge to urinate.

🩸 More to know: artificial kidney (haemodialysis)

- Infections, injury or restricted blood flow can reduce kidney activity; poisonous wastes build up and can cause **death**.
- An **artificial kidney** removes nitrogenous wastes from blood by **dialysis**.
- It has tubes with a **semi-permeable** lining in a tank of **dialysing fluid**, which has the **same osmotic pressure as blood** but **no nitrogenous wastes**.
- Blood passes through the tubes; wastes **diffuse** into the fluid; purified blood is pumped back.
- Difference from a kidney: **no re-absorption**. A healthy kidney filters about **180 L** daily but only **1–2 L** is excreted.



Artificial kidney (from the More to know box, page 97)

🤝 Think it over: organ donation

Donating an organ to a person whose organ has failed is a generous act, done with the consent of the donor and family. Anyone, of any age or gender, can be a donor. Common transplants: **corneas, kidneys, heart, liver, pancreas, lungs, intestines, bone marrow**. Most donations happen after death or brain death, but a **kidney, part of a liver, lung** and some tissues can be donated while alive.

5.5.2 Excretion in plants

Plants use completely different strategies:

🌬️ Gases

Oxygen from photosynthesis is itself a waste product; O_2 and CO_2 leave by diffusion through stomata.

💦 Excess water

Removed by **transpiration**.

🍂 Falling leaves

Wastes are stored in leaves that **fall off**; plants can lose parts because many tissues are **dead cells**.

🫙 **Vacuoles**
Many wastes are stored in **cellular vacuoles**.

🌲 **Resins and gums**
Stored especially in **old xylem**.

🌍 **Into the soil**
Some wastes are excreted into the **soil** around the roots.



Differences to learn (2- and 3-mark favourites)

Autotrophic vs heterotrophic nutrition

Autotrophic	Heterotrophic
Organism makes its own food	Organism depends on food made by others
Raw materials: simple inorganic (CO_2, water)	Takes in complex organic substances
Needs sunlight and chlorophyll	Needs enzymes to break food down
Green plants, some bacteria	Animals, fungi

Aerobic vs anaerobic respiration

Aerobic	Anaerobic
Uses oxygen	Without oxygen
Pyruvate broken down in **mitochondria**	Pyruvate broken down in **cytoplasm**
Products: CO_2 + water + energy	Ethanol + CO_2 (yeast) or lactic acid (muscle) + energy
Much more energy released	Much less energy
Humans, most plants and animals	Yeast, some bacteria, our muscle cells during heavy exercise

Breathing vs respiration

Breathing	(Cellular) respiration
Physical process: taking air in and out of the lungs	Chemical process: breaking down food to release energy
Happens in the respiratory organs	Happens in every cell (cytoplasm and mitochondria)
No energy is released	Energy released and stored as ATP

Xylem vs phloem (NCERT Exercise 12)

Xylem	Phloem
Carries **water and minerals**	Carries **food** (sugars, amino acids)
From roots to leaves: **upward only**	From leaves to other parts: **both upward and downward**
Explained largely by **physical forces** (root pressure, transpiration pull); no energy used	Uses **energy from ATP** (osmotic pressure)
Through vessels and tracheids (mostly dead cells)	Through sieve tubes with companion cells (living)

Transpiration vs translocation

Transpiration	Translocation
Loss of water vapour from aerial parts	Transport of soluble food in the plant
Through stomata (mainly)	Through phloem
Creates the pull for water in xylem; cools the plant	Supplies food to storage and growing organs

Blood vs lymph

Blood	Lymph
Red (has RBCs with haemoglobin)	Colourless
More protein	Less protein
Flows in arteries, veins, capillaries, pumped by the heart	Flows in lymph vessels, drains into large veins
Carries oxygen, food, CO_2, wastes	Carries digested fat; drains excess tissue fluid

Alveoli vs nephrons (NCERT Exercise 13)

	Alveoli (lungs)	Nephrons (kidneys)
Structure	Tiny balloon-like sacs at the ends of bronchioles	Coiled tubes with a cup-shaped Bowman's capsule at one end
Capillaries	Network of capillaries in the alveolar walls	Glomerulus (capillary cluster) in Bowman's capsule, capillaries around the tubule

	Alveoli (lungs)	Nephrons (kidneys)
Number	Millions; surface about 80 m^2	Large numbers in each kidney
Function	Exchange of gases: O_2 into blood, CO_2 out	Filter nitrogenous waste (urea) from blood and re-absorb useful substances
Process	Diffusion	Filtration and selective re-absorption
Common feature	Both have very thin walls, a huge surface area and a rich supply of blood capillaries.	

Nephron vs neuron (CBSE 2026)

Nephron	Neuron
Structural and functional unit of the **kidney**	Structural and functional unit of the **nervous system**
Filters blood to form urine	Carries messages (nerve impulses)

Haemodialysis vs natural kidney

Artificial kidney	Natural kidney
Wastes removed by diffusion through a semi-permeable tube	Filtration in the glomerulus
No re-absorption	Selective re-absorption of glucose, amino acids, salts, water
Machine, used outside the body for some hours	Works all the time inside the body

Draw

Diagrams for the exam

Tap any diagram to open it full size. Practise drawing each one with all its labels; board papers often ask you to draw and label one of these for 2 to 5 marks.

✏️ Drawing tips that save marks

- Use a pencil, draw label lines with a ruler, keep labels on one side and do not let lines cross.
- Heart: draw the left ventricle wall the thickest; show the septum; label all four chambers and four vessels (aorta, pulmonary artery, pulmonary veins, vena cava).
- Nephron: glomerulus inside Bowman's capsule, coiled tubule, capillaries around it, collecting duct.
- Write the figure title below the diagram.

CBSE Previous year board questions

These questions come from the CBSE Class 10 Science board papers of the years shown, with the paper code (set) and marks. Tap a question to see a model answer.

Where we could not check a question word for word (for example its options or its case text), it is **reworded** and marked so. The years, sets and marks shown are only those we could confirm from the paper or a published solution of it.

– 2026 · 31/1/1 · 1 mark

Which structure in a leaf is mainly responsible for gaseous exchange? (a) Xylem (b) Stomata (c) Phloem (d) Cuticle

(b) Stomata. Massive gas exchange for photosynthesis takes place through these pores; xylem and phloem conduct water and food, and the cuticle is a waxy waterproof layer.

– 2026 · 31/1/1 · 1 mark

Pancreas secretes pancreatic juice which contains certain enzymes that help in digestion of food. Which is correct? (a) Trypsin digests emulsified fats and lipase digests proteins. (b) Trypsin digests proteins and lipase digests emulsified fats. (c) Trypsin and lipase both digest fats. (d) Trypsin digests proteins and lipase digests carbohydrates.

(b) Trypsin digests proteins; lipase breaks down emulsified fats.

– 2026 · 31/1/1 · 1 mark

Assertion (A): In human beings, the respiratory pigment is haemoglobin present in red blood cells. Reason (R): Haemoglobin has a very high affinity for carbon dioxide.

(c) A is true, but R is false. Haemoglobin has a very high affinity for **oxygen**. CO_2 is mostly carried dissolved in plasma.

– 2026 · 31/1/1 · 1 + 1 marks

(i) What is the function of the diaphragm in the human respiratory system? (ii) Where is the diaphragm present in the human body?

(i) When we breathe in, the diaphragm **flattens** (and the ribs are lifted), so the **chest cavity becomes larger** and air is sucked into the lungs; when it relaxes back into a dome, air is pushed out. (ii) It is a dome-shaped muscular sheet **below the lungs**, separating the **chest (thoracic) cavity** from the **abdominal cavity**.

– 2026 · 31/1/1 · 2 marks

State two differences between the act of chewing food and salivation on sight of food.

(1) **Chewing** is a **voluntary** action, done consciously and controlled by the forebrain (cerebrum); **salivation** on seeing food is an **involuntary** action that happens on its own, controlled by the **medulla** in the hind-brain. (2) Chewing is the **mechanical breakdown** of food by the teeth and tongue (muscle action); salivation is the **secretion of saliva** by the salivary glands, which wets the food and starts starch digestion with salivary amylase.

– 2026 · 31/1/1 · 1 mark **State the difference between nephron and neuron.**

A **nephron** is the structural and functional unit of the **kidney** that filters blood to form urine; a **neuron** is the structural and functional unit of the **nervous system** that carries nerve impulses.

– 2025 · 31/1/1 · 1 mark

Secretion of less saliva in mouth will affect the conversion of: (A) proteins into amino acids (B) fats into fatty acids and glycerol (C) starch into simple sugars (D) sugars into alcohol

(C) starch into simple sugars. Saliva contains salivary amylase, which digests starch.

– 2025 · 31/1/1 · 1 mark (reworded)

Glucose → (site 'a') → pyruvate + energy → (site 'b') → lactic acid + energy. The sites 'a' and 'b' respectively are: (A) Mitochondria and oxygen deficient muscle cells (B) Cytoplasm and oxygen rich muscle cells (C) Cytoplasm and yeast cells (D) Cytoplasm and oxygen deficient muscle cells

(D) Cytoplasm and oxygen deficient muscle cells. Glucose → pyruvate always happens in the cytoplasm; lactic acid forms when muscle cells lack oxygen. (The flow chart in the paper is described in words here.)

– 2025 · 31/1/1 · 2 marks

(a) Besides minimising the loss of blood, why is it essential to plug any leak in a blood vessel? Name the component of blood that helps in this and state how. OR (b) (i) Plants have a transport system that is relatively slower than in animals. Give reasons. (ii) State the role of phloem in the transport of materials in plants.

(a) A leak would cause a **loss of pressure**, which reduces the **efficiency of the pumping system** (the heart). **Platelets** help: they circulate in the blood and plug the leak by helping the **blood to clot** at the injury.
OR (b) (i) Plants do not move, and a large proportion of their tissues are **dead cells**, so they have **low energy needs** and a slow transport system is enough. (ii) Phloem carries **soluble products of photosynthesis** (sugars) and amino acids from the leaves to other parts, especially **storage organs** (roots, fruits, seeds) and growing parts, both upward and downward.

2025 · 31/1/1, Q38, case-based, 4 marks. Case (reworded): the maintenance functions of living organisms must go on even when they are not doing anything; autotrophs (producers) make the food that supplies energy to all organisms.

– 2025 · 1 mark **(a) Name and define the process by which green plants prepare food.**

Photosynthesis: the process by which green plants use **sunlight** and **chlorophyll** to convert **carbon dioxide and water** into **carbohydrates** (glucose), releasing oxygen.

– 2025 · 1 mark **(b) Write the chemical equation involved in the above process.**

$6CO_2 + 12H_2O \xrightarrow{\text{(chlorophyll, sunlight)}} C_6H_{12}O_6 + 6O_2 + 6H_2O$

– 2025 · 2 marks

(c) (i) State in proper sequence the events that occur in the synthesis of food by desert plants.

Desert plants **take up CO_2 at night** and make an **intermediate** compound. During the day: (1) **absorption of light energy** by chlorophyll; (2) **conversion of light energy to chemical energy** and **splitting of water** into hydrogen and oxygen; (3) **reduction of CO_2** (from the intermediate) to carbohydrates.

– 2025 · 2 marks (OR)

(c) (ii) Explain, giving reasons, what happens to the rate at which green plants prepare food (I) during cloudy weather (II) when stomata get blocked due to dust.

(I) The rate **falls**: less **sunlight** reaches the leaves, so less light energy is absorbed by chlorophyll. (II) The rate **falls**: blocked stomata reduce the **intake of CO_2** (and release of O_2), and CO_2 is a raw material for photosynthesis.

– 2024 · 31/1/1 · 2 marks (reworded)

State the role of (i) hydrochloric acid in the stomach (ii) villi in the small intestine (iii) anal sphincter (iv) lipase.

(i) **HCl** creates an **acidic medium** that helps the enzyme pepsin act (and kills germs). (ii) **Villi increase the surface area** for absorption of digested food. (iii) The **anal sphincter regulates the exit** of waste material through the anus. (iv) **Lipase** breaks down **emulsified fats** into fatty acids and glycerol.

– 2024 · 31/1/1 · 3 marks (reworded)

(i) Why do large animals like humans need a respiratory pigment? (ii) Give reasons: (a) rings of cartilage are present in the throat; (b) the lungs always contain a residual volume of air; (c) the ribs are lifted and the diaphragm flattened when we breathe in; (d) the alveoli are supplied with blood vessels.

(i) In large animals, **diffusion pressure alone cannot deliver oxygen** to all parts of the body, so a pigment (haemoglobin) carries oxygen from the lungs to the tissues. (ii) (a) They ensure the **air passage does not collapse**. (b) There is **enough time for O_2 to be absorbed** and CO_2 released. (c) The **chest cavity becomes larger**, so air is sucked in. (d) **Exchange of gases** between alveolar air and blood takes place there.

The next four are from the CBSE Class 10 Science paper of 2 March 2024 (a different set from 31/1/1; set code not confirmed).

– 2024 · 1 mark

When digestion is complete, proteins, carbohydrates and fats are converted respectively into: (a) amino acids, glucose, fatty acids (b) amino acids, glucose, fatty acids and glycerol (c) glucose, fatty acids and glycerol, amino acids (d) sugars, amino acids, fatty acids and glycerol

(b) Proteins → amino acids; complex carbohydrates → glucose; fats → fatty acids and glycerol.

– 2024 · 1 mark (reworded)

Assertion (A): Most plants close their stomata at night. Reason (R): Closing the stomata conserves water, since large amounts of water would otherwise be lost from the leaves.

(a) Both A and R are true, and R explains A. At night there is no photosynthesis, so CO_2 is not needed; the plant closes its stomata to avoid losing water.

– 2024 · 2 marks

Explain how water applied to the soil reaches the leaves of a plant. OR Name the type of nutrition in Amoeba, and explain how it takes in and digests its food.

Root cells take up **ions** from the soil actively; the concentration difference makes **water enter the root xylem** and pushes a column of water up (**root pressure**). Xylem vessels and tracheids form a **continuous channel** to the leaves. **Transpiration** from leaves creates a **suction (transpiration pull)** that pulls water up to the leaves; this is the main force by day.
OR Amoeba shows **heterotrophic (holozoic)** nutrition. It puts out **pseudopodia** that surround the food and fuse to form a **food vacuole**; enzymes break the food down inside the vacuole, and the simple substances **diffuse into the cytoplasm**; undigested matter is thrown out at the surface.

2024, case-based, 4 marks (reworded). Case: kidneys are vital organs; infections, injury or restricted blood flow reduce their activity, wastes accumulate and this can cause death; in kidney failure an artificial kidney removes wastes by dialysis.

– 2024 · case

(i) Name (a) the artery that brings oxygenated blood to the kidney (b) the cluster of thin-walled capillaries inside Bowman's capsule.

(a) **Renal artery**. (b) **Glomerulus**.

– 2024 · case **(ii) Name the organ that stores urine. Is it under hormonal or nervous control?**

Urinary bladder. It is muscular and under **nervous control**, so we can usually control the urge to urinate.

– 2024 · case

(iii) List the two major steps of urine formation and state their functions. OR Where in the nephron does selective re-absorption occur, and what decides the amount of water re-absorbed?

(1) **Filtration** in the glomerulus and Bowman's capsule: removes water, urea, salts, glucose etc. from blood into the capsule as filtrate. (2) **Selective re-absorption** in the tubular part: useful substances (glucose, amino acids, salts, much of the water) are taken back into the blood, leaving urine.

OR In the **tubular part** of the nephron. The amount of water re-absorbed depends on **how much excess water** there is in the body and **how much dissolved waste** has to be excreted.

– 2023 · 31/1/1 · 2 marks (reworded)

Name the blood vessel that (i) brings deoxygenated blood from the body to the heart (ii) carries deoxygenated blood from the heart to the lungs. (The other alternative of this question was based on a diagram of the heart.)

(i) **Vena cava**, into the right atrium. (ii) **Pulmonary artery**, from the right ventricle.

– 2023 · 31/1/1 · 2 marks (reworded) **Describe the structure of a nephron and state its function.**

Structure: a cluster of thin-walled capillaries, the **glomerulus**, sits in the cup-shaped end of a coiled tube called **Bowman's capsule**; the tube continues as the tubular part and joins a collecting duct. **Function: filtration** of nitrogenous wastes from blood to form urine (also re-absorption of useful substances, osmoregulation).

– 2023 · 31/1/1 · 3 marks (reworded)

Part (a): (i) Why is ATP called the energy currency of the cell? (ii) Through which parts do plants exchange gases? (iii) What decides the direction of diffusion of gases in plants? Part (b): (i) In what form is food stored in plants and in animals? (ii) How do desert plants carry out photosynthesis?

(a) (i) Energy from respiration is stored as ATP, and ATP is used to fuel **all other cellular activities** (muscle contraction, protein synthesis, nerve impulses), so it is the **energy currency for cellular processes**. (ii) Through **stomata**, and across the surface of **leaves, stems and roots**. (iii) The **environmental conditions** and the **requirements of the plant** (e.g. day: O_2 out; night: CO_2 out). (b) (i) Plants: **starch**; animals: **glycogen**. (ii) Desert plants **take up CO_2 at night** and make an **intermediate**; during the day the energy absorbed by chlorophyll acts on this intermediate to make carbohydrates. This saves water, as stomata stay closed in the hot day.

Board-pattern practice

Extra questions in the current CBSE pattern (not from a specific year). Tap to check.

Assertion–Reason (1 mark each)

(a) Both A and R are true, and R explains A. (b) Both true, but R does not explain A. (c) A true, R false. (d) A false, R true.

– **1** **A: Ventricles have thicker walls than atria. R: Ventricles pump blood to various organs of the body.**

(a)

– **2** **A: Herbivores have a longer small intestine than carnivores. R: Cellulose takes longer to digest.**

(a)

– **3** **A: The rate of breathing in aquatic organisms is faster than in terrestrial organisms. R: Water has more dissolved oxygen than air.**

(c) Dissolved oxygen in water is much **lower** than in air.

– **4** **A: Bile juice contains no enzyme but is important for digestion. R: Bile salts emulsify fats and make the food alkaline.**

(a)

– **5** **A: Translocation in phloem needs energy. R: Sucrose is loaded into phloem using ATP.**

(a)

– **6** **A: Veins have valves. R: Blood in veins flows under very high pressure.**

(c) Blood in veins is under **low** pressure; valves keep it flowing one way.

Short answer

S1 Why do muscles get cramps after a sprint?

During sudden activity muscle cells run short of oxygen, so pyruvate is converted into **lactic acid** (anaerobic pathway). The build-up of lactic acid causes cramps.

S2 Why is the left ventricle wall the thickest of all chambers?

It pumps oxygenated blood through the aorta to the **whole body**, which needs the greatest force; the right ventricle only pumps to the nearby lungs.

S3 Why is the mucus secreted by gastric glands important?

It **protects the inner lining** of the stomach from hydrochloric acid (and pepsin). Without it the stomach wall would be damaged, causing ulcers.

S4 A plant is kept in the dark for three days before an experiment on photosynthesis. Why?

To **destarch** the leaves: all the stored starch is used up, so any starch found later must have been made during the experiment.

S5 Why do desert plants open stomata at night?

Opening stomata in the hot day would lose a lot of **water**. At night it is cooler, so they take in CO_2 then, store it as an intermediate and use it in the day.

S6 Out of 180 L of filtrate per day, only about 1.5 L of urine is passed. What percentage is re-absorbed?

(180 − 1.5) / 180 × 100 ≈ **99.2%**.

Case-based (4 marks)

– **C1** **Riya ran 400 m fast and felt pain in her legs. Her teacher said the cells switched pathway. (i) Name the substance that built up. (ii) Where in the cell does glucose break into pyruvate? (iii) Write the pathway when oxygen is present, naming the organelle. (iv) Which pathway releases more energy and why does the body prefer it?**

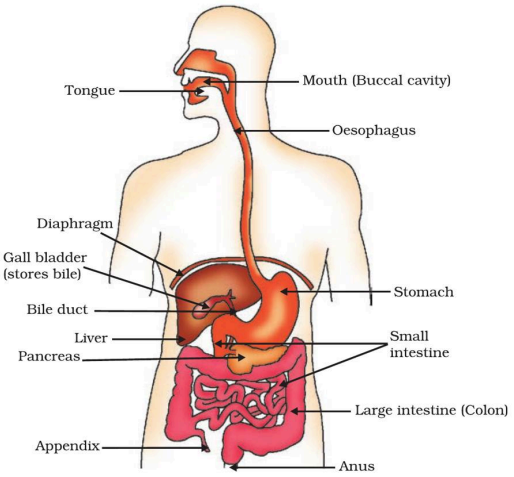
(i) **Lactic acid**. (ii) **Cytoplasm**. (iii) Glucose → pyruvate (cytoplasm) → CO_2 + water + energy in the **mitochondria**. (iv) **Aerobic**: glucose is broken down completely, so much more energy (ATP) is released; anaerobic breakdown is incomplete.

– **C2** **A student kept a potted plant with a leaf partly covered by black paper strips in sunlight for 6 hours (after destarching) and then tested the leaf with iodine. (i) Which part turns blue-black? (ii) What does this prove? (iii) Why is the leaf boiled in alcohol? (iv) Why on a water-bath?**

(i) The **uncovered** part. (ii) **Sunlight is essential** for photosynthesis. (iii) To **remove chlorophyll**, so the iodine colour can be seen clearly. (iv) Alcohol is **inflammable**; direct heating could set it on fire.

Long answer (5 marks)

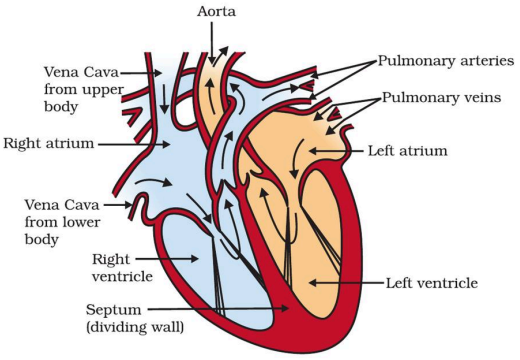
– **L1** **Draw a labelled diagram of the human alimentary canal. Describe the role of the liver, pancreas and small intestine in digestion.**



Model diagram: Fig. 5.6

Liver: makes **bile** (stored in the gall bladder), which makes the acidic food alkaline and **emulsifies fats**. **Pancreas**: pancreatic juice with **trypsin** (proteins) and **lipase** (emulsified fats). **Small intestine**: site of **complete digestion**; intestinal juice converts proteins → amino acids, carbohydrates → glucose, fats → fatty acids and glycerol; **villi** absorb digested food into the blood.

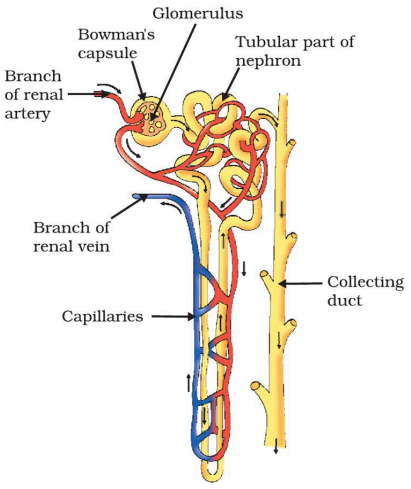
– **L2 Describe the structure of the human heart with a diagram, and explain double circulation. Why is it necessary?**



Model diagram: Fig. 5.10

Four chambers: two thin-walled atria above, two thick-walled ventricles below; septum separates right and left; valves prevent backflow. Right side handles deoxygenated blood (vena cava → RA → RV → pulmonary artery → lungs); left side handles oxygenated blood (pulmonary veins → LA → LV → aorta → body). Blood passes through the heart **twice** per cycle: double circulation. It keeps oxygenated and deoxygenated blood separate, giving a highly efficient oxygen supply needed by warm-blooded birds and mammals with high energy needs.

– **L3** **Draw the structure of a nephron and explain how urine is formed. How is the amount of urine regulated?**



Model diagram: Fig. 5.14

Blood enters the glomerulus by a branch of the renal artery; water, urea, salts, glucose and amino acids are **filtered** into Bowman's capsule. As the filtrate flows along the tubule, glucose, amino acids, salts and most water are **selectively re-absorbed** into surrounding capillaries. The remaining fluid is urine, which goes to the collecting duct, ureter and bladder. Amount of water re-absorbed depends on excess water in the body and the amount of dissolved waste to be excreted.

Recap

What you have learnt

- Movement of various types (including molecular movement) is an indication of life.
- Maintenance of life needs **nutrition, respiration, transport** of materials and **excretion** of wastes.
- **Autotrophic** nutrition: simple inorganic materials + an external energy source (Sun) → complex high-energy organic material (photosynthesis, needs chlorophyll, CO_2, water, sunlight).
- **Heterotrophic** nutrition: intake of complex material made by other organisms.
- In humans, food is broken down step by step along the alimentary canal and absorbed in the **small intestine** (villi).
- Respiration breaks down glucose to give energy as **ATP**. **Aerobic** respiration (mitochondria) gives much more energy than **anaerobic**.
- The circulatory system (**heart, blood, blood vessels**) transports oxygen, CO_2, food and wastes. Humans have **double circulation**.
- In plants, **xylem** carries water and minerals (root pressure, transpiration pull) and **phloem** carries food (translocation, uses energy).

- In humans, soluble nitrogenous wastes are removed by **nephrons** in the kidneys.
- Plants remove wastes by storing them in vacuoles, as gums and resins, in falling leaves, or by excreting them into the soil.

Quiz Test yourself

Tap an answer. **Score: 0 / 20**

1. The kidneys in human beings are a part of the system for (NCERT Ex. 1)

Nutrition	Respiration
Excretion	Transportation

Kidneys filter nitrogenous waste from blood.

2. The xylem in plants is responsible for (NCERT Ex. 2)

Transport of water	Transport of food
Transport of amino acids	Transport of oxygen

Food and amino acids move in phloem.

3. The autotrophic mode of nutrition requires (NCERT Ex. 3)

CO_2 and water	Chlorophyll
Sunlight	**All of the above**

All three are needed for photosynthesis.

4. Breakdown of pyruvate to give CO_2, water and energy takes place in (NCERT Ex. 4)

- Cytoplasm
- **Mitochondria**
- Chloroplast
- Nucleus

Aerobic breakdown of pyruvate happens in mitochondria.

5. Salivary amylase acts on

- Proteins
- **Starch**
- Fats
- Vitamins

Starch → simple sugar.

6. Which juice has no enzyme but helps digest fats?

- Saliva
- Gastric juice
- **Bile**
- Pancreatic juice

Bile salts emulsify fats.

7. Pepsin works best in a medium that is

- **Acidic**
- Alkaline
- Neutral
- Any

HCl makes the stomach acidic for pepsin.

8. Finger-like projections that increase absorption in the small intestine are

- Alveoli
- **Villi**
- Cilia
- Nephrons

Alveoli are in lungs, cilia line the air passage.

9. Yeast in the absence of oxygen produces

- Lactic acid + energy
- **Ethanol + CO_2 + energy**
- CO_2 + water + energy
- Glucose

Fermentation.

10. Muscle cramps after heavy exercise are due to

- Ethanol
- Pyruvate
- **Lactic acid**
- Glucose

Pyruvate → lactic acid when oxygen is short.

11. The energy currency of the cell is

- Glucose
- ADP
- **ATP**
- Starch

Breaking its terminal phosphate bond gives 30.5 kJ/mol.

12. Rings of cartilage in the throat

- Filter dust
- **Stop the air passage from collapsing**
- Make mucus
- Exchange gases

Hairs and mucus filter the air.

13. Haemoglobin is found in

- Plasma
- Platelets
- **Red blood corpuscles**
- Lymph

It has a high affinity for oxygen.

14. Which vessel carries oxygenated blood to the heart?

- Pulmonary artery
- **Pulmonary vein**
- Vena cava
- Aorta

From lungs to the left atrium.

15. Fishes have a heart with

- **2 chambers**
- 3 chambers
- 4 chambers
- No chambers

Single circulation through the gills.

16. Normal blood pressure of an adult is about

- 80/120 mm Hg
- **120/80 mm Hg**
- 140/100 mm Hg
- 100/60 kPa

Systolic 120, diastolic 80.

17. During the day, the main force moving water up in xylem is

- Root pressure
- **Transpiration pull**
- Gravity
- Phloem pressure

Root pressure matters more at night.

18. Translocation of food takes place in

- Xylem vessels
- Tracheids
- **Sieve tubes**
- Stomata

With help from companion cells.

19. The cluster of capillaries in Bowman's capsule is the

- **Glomerulus**
- Alveolus
- Ureter
- Villus

Filtration happens here.

20. Daily initial filtrate in the kidneys of a healthy adult is about

1–2 L	18 L
180 L	1800 L

Only 1–2 L is passed as urine; the rest is re-absorbed.

Answers to all questions

Every question from the chapter, with answers you can write in the exam. Tap to open.

Questions (page 81)

1. Why is diffusion insufficient to meet the oxygen requirements of multicellular organisms like humans?

In multicellular organisms like humans, the body is large and **most cells are not in direct contact** with the air. Diffusion is very **slow over long distances** (oxygen would take about 3 years to reach the toes from the lungs), so it cannot supply oxygen fast enough to all cells. Humans also have **high energy needs**. So they need special organs (lungs), a pigment (haemoglobin) and a transport system (heart and blood).

2. What criteria do we use to decide whether something is alive?

We look for **movement** of some kind: visible movements like breathing, walking or growth, and, more fundamentally, **molecular movement** inside the body. Living things carry out **life processes** (nutrition, respiration, transport, excretion) to maintain and repair their organised structure. Something showing no molecular movement (like a virus outside a cell) is not clearly alive.

3. What are outside raw materials used for by an organism?

(1) As a **source of energy** (food is broken down in respiration to give energy for life processes). (2) As **raw material** for **growth**, repair and making new substances such as proteins. Examples: food, water, oxygen, CO_2 (for plants), minerals.

– **4. What processes would you consider essential for maintaining life?**

Nutrition, respiration, transportation and excretion. (Also growth, reproduction, and control and coordination help keep the organism alive and its species going.)

Questions (page 87)

– **1. What are the differences between autotrophic nutrition and heterotrophic nutrition?**

See the table. Autotrophic: organism makes its own food from simple inorganic CO_2 and water using sunlight and chlorophyll (green plants, some bacteria). Heterotrophic: organism takes in complex food made by others and breaks it down with enzymes (animals, fungi); depends on autotrophs.

– **2. Where do plants get each of the raw materials required for photosynthesis?**

Carbon dioxide: from the air, through stomata. **Water**: from the soil, absorbed by roots and carried by xylem. **Sunlight**: from the Sun. **Chlorophyll**: present in chloroplasts of leaf cells. (Minerals like nitrogen and magnesium come from the soil.)

– **3. What is the role of the acid in our stomach?**

Hydrochloric acid creates an **acidic medium** that helps the enzyme **pepsin** digest proteins. It also **kills germs** that enter with food.

– **4. What is the function of digestive enzymes?**

Digestive enzymes are biological catalysts that **break down complex food molecules into simpler, soluble ones** that can be absorbed: starch → sugar (amylase), proteins → amino acids (pepsin, trypsin), fats → fatty acids and glycerol (lipase).

– **5. How is the small intestine designed to absorb digested food?**

It is very **long and coiled**, and its inner lining has numerous finger-like projections called **villi**, which greatly **increase the surface area** for absorption. The villi are **richly supplied with blood vessels** that carry absorbed food to every cell (fat is absorbed into lymph).

Questions (page 91)

1. What advantage over an aquatic organism does a terrestrial organism have with regard to obtaining oxygen for respiration?

Air has a **much higher oxygen content** than water (in which oxygen is only dissolved in small amounts). So a terrestrial organism gets enough oxygen with a **slower breathing rate** and less effort, while aquatic organisms must breathe much faster.

2. What are the different ways in which glucose is oxidised to provide energy in various organisms?

First, in all organisms, glucose (6-C) → pyruvate (3-C) + energy in the **cytoplasm**. Then: (1) **Absence of oxygen (yeast)**: pyruvate → ethanol + CO_2 + energy (fermentation). (2) **Lack of oxygen (our muscles)**: pyruvate → lactic acid + energy. (3) **Presence of oxygen (mitochondria)**: pyruvate → CO_2 + water + much more energy. See Fig. 5.8.

3. How are oxygen and carbon dioxide transported in human beings?

Oxygen is carried by the respiratory pigment **haemoglobin** in **red blood corpuscles**, which has a very high affinity for oxygen; it picks up O_2 in the lungs and releases it in tissues. **Carbon dioxide** is more soluble in water, so it is mostly carried **dissolved in the plasma**.

4. How are the lungs designed in human beings to maximise the area for exchange of gases?

In the lungs the air passage divides into smaller and smaller tubes (bronchi → bronchioles) that end in millions of balloon-like **alveoli**. If spread out, the alveolar surface would cover about **80 m^2**. Alveolar walls are **very thin** and have an **extensive network of capillaries**, so gases diffuse quickly.

Questions (page 96)

1. What are the components of the transport system in human beings? What are the functions of these components?

(1) **Heart**: pumps blood around the body. (2) **Blood vessels**: arteries carry blood away from the heart, veins bring it back, capillaries exchange materials with cells. (3) **Blood**: plasma carries food, CO_2, nitrogenous wastes and salts; RBCs carry oxygen; **platelets** clot blood to plug leaks. (4) **Lymph**: carries digested fat from the intestine and drains excess tissue fluid back into blood.

2. Why is it necessary to separate oxygenated and deoxygenated blood in mammals and birds?

Mammals and birds have **high energy needs** because they constantly use energy to **maintain their body temperature**. Keeping oxygenated and deoxygenated blood separate gives a **highly efficient supply of oxygen** to the body cells, so they can release more energy.

3. What are the components of the transport system in highly organised plants?

The vascular tissue: **xylem** (vessels and tracheids), which carries water and minerals from roots to leaves, and **phloem** (sieve tubes and companion cells), which carries food from leaves to other parts.

4. How are water and minerals transported in plants?

Through **xylem**, a continuous system of vessels and tracheids from roots to leaves. Root cells actively take up ions, so water moves into the root and a column of water is pushed up (**root pressure**, important at night). In the day, **evaporation of water from leaves (transpiration)** creates a suction, the **transpiration pull**, which is the main force pulling water and dissolved minerals up to the top.

5. How is food transported in plants?

By **translocation** in the **phloem**, through sieve tubes helped by companion cells, upwards and downwards. Sucrose is loaded into phloem using energy from **ATP**; this raises the **osmotic pressure**, so water enters and the pressure pushes the material to tissues with lower pressure, such as storage organs, fruits, seeds and growing buds.

Questions (page 98)

1. Describe the structure and functioning of nephrons.

Structure: each nephron has a cup-shaped **Bowman's capsule** holding a cluster of thin-walled capillaries, the **glomerulus**, followed by a long coiled **tubule** surrounded by capillaries, which opens into a collecting duct. **Functioning:** (1) **Filtration**: blood from the renal artery is filtered in the glomerulus; water, urea, salts, glucose and amino acids pass into the capsule as filtrate. (2) **Selective re-absorption**: in the tubule, glucose, amino acids, salts and much water are taken back into blood. (3) What remains is **urine**, which goes to the collecting duct and ureter.

2. What are the methods used by plants to get rid of excretory products?

(1) O_2 and CO_2 by diffusion through stomata. (2) Excess water by **transpiration**. (3) Storing wastes in **leaves that fall off** (and bark). (4) Storing wastes in **cell vacuoles**. (5) As **gums and resins**, especially in old xylem. (6) Excreting some wastes into the **soil**.

3. How is the amount of urine produced regulated?

By controlling how much water is **re-absorbed** in the kidney tubules. This depends on (1) how much **excess water** there is in the body and (2) how much **dissolved waste** must be excreted. More water in the body → less re-absorbed → more urine; less water (hot day, sweating) → more re-absorbed → less, concentrated urine. (Hormones control this; you will meet them in higher classes.)

Activities

5.1 Variegated leaf starch test: what is the colour of the leaf and of the alcohol after boiling? What do you conclude?

Leaf becomes **pale/whitish**; alcohol turns **green** (chlorophyll dissolves). With iodine, only the earlier **green** areas turn **blue-black**: starch is made only where chlorophyll is present. See section 5.2.1.

5.2 KOH bell-jar experiment: do both leaves have the same starch? Conclusion? Design an experiment to show sunlight is essential.

No: the leaf from the jar with KOH has **no starch**; the other has starch. So **CO_2 is essential**. For sunlight: destarch a plant, cover part of a leaf on both sides with **black paper** strips, keep it in sunlight for 6 hours, then do the starch test. Only the **uncovered** part turns blue-black, so sunlight is essential. Try it in the photosynthesis lab.

5.3 Starch and saliva: which tube changes colour?

Tube B (no saliva) turns blue-black; tube A does not, since saliva has digested the starch. See section 5.2.4.

5.4 Lime water with exhaled air and with ordinary air.

Exhaled air turns lime water milky much faster: exhaled air has more CO_2.

5.5 Yeast and sugar solution with lime water.

Lime water turns milky: fermentation produces CO_2 (and ethanol).

5.6 Watching fish breathe: are mouth and gill-slit movements coordinated? Compare rates.

Yes: when the fish **opens its mouth**, the gill-slits (operculum) **close**; when it closes its mouth, the gill-slits **open** and water flows out over the gills. The fish's rate is **much higher** than our breathing rate (about 12–20 per minute at rest) because water contains much less dissolved oxygen. Use the breath counter.

5.7 Haemoglobin levels in humans and animals.

See the answer in section 5.4.1.

5.8 Plant and stick under plastic sheets.

Droplets form only inside the plastic over the plant: water is lost from leaves by transpiration.

In text What other function do you think is served by the acid in the stomach? Can 'acidity' be related to this?

The acid **kills bacteria and germs** that enter with food. "Acidity" is the discomfort or burning felt when the stomach releases **too much acid** (or the acid reaches the food-pipe).

Exercises (page 99)

1. The kidneys in human beings are a part of the system for (a) nutrition (b) respiration (c) excretion (d) transportation

(c) excretion.

2. The xylem in plants are responsible for (a) transport of water (b) transport of food (c) transport of amino acids (d) transport of oxygen

(a) transport of water.

3. The autotrophic mode of nutrition requires (a) carbon dioxide and water (b) chlorophyll (c) sunlight (d) all of the above

(d) all of the above.

4. **The breakdown of pyruvate to give carbon dioxide, water and energy takes place in (a) cytoplasm (b) mitochondria (c) chloroplast (d) nucleus**

(b) mitochondria.

5. **How are fats digested in our bodies? Where does this process take place?**

Fat digestion takes place in the **small intestine**. Fats are present as large globules, so enzymes cannot act easily. **Bile** from the liver makes the medium alkaline, and **bile salts emulsify** the fats into small globules. **Lipase** in pancreatic juice breaks down the emulsified fats, and enzymes of the **intestinal juice** finally convert them into **fatty acids and glycerol**, which are absorbed by the villi.

6. **What is the role of saliva in the digestion of food?**

Saliva (1) **wets the food** so it is easy to chew and swallow and passes smoothly through the canal, and (2) contains the enzyme **salivary amylase**, which breaks down **starch** into **simple sugar**.

7. **What are the necessary conditions for autotrophic nutrition and what are its by-products?**

Conditions: sunlight, chlorophyll, carbon dioxide and water (and a suitable temperature). **By-products: oxygen** (and water). The main product is carbohydrate (glucose), stored as starch.

8. **What are the differences between aerobic and anaerobic respiration? Name some organisms that use the anaerobic mode of respiration.**

See the table: aerobic uses oxygen, completes in mitochondria, gives CO_2 + water and much more energy; anaerobic needs no oxygen, happens in cytoplasm, gives ethanol + CO_2 or lactic acid and less energy. **Organisms: yeast**, some **bacteria** (e.g. those that make curd use lactic acid fermentation), and **our muscle cells** during heavy exercise. Tapeworms living in the gut also respire anaerobically.

9. **How are the alveoli designed to maximise the exchange of gases?**

(1) There are millions of alveoli, giving a **huge surface area** (about 80 m^2). (2) Their walls are **very thin** (one cell thick), so gases diffuse quickly. (3) They have an **extensive network of blood capillaries**. (4) The lungs keep a **residual volume** of air, giving enough time for exchange. (5) They are moist and kept inside the body for protection.

10. What would be the consequences of a deficiency of haemoglobin in our bodies?

Less oxygen would be carried to the cells, so less energy would be released. The person would feel **tired, weak, breathless** and look pale, with poor ability to work. This condition is called **anaemia**.

11. Describe double circulation of blood in human beings. Why is it necessary?

Blood passes through the heart **twice** in one complete round. (1) **Pulmonary circulation**: deoxygenated blood from the body (vena cava) → right atrium → right ventricle → pulmonary artery → lungs, where it is oxygenated → pulmonary veins → left atrium. (2) **Systemic circulation**: left atrium → left ventricle → aorta → body organs, where oxygen is used → vena cava → right atrium. **Necessity:** it keeps oxygenated and deoxygenated blood separate, giving a highly efficient supply of oxygen, which humans need for their high energy demand and to maintain constant body temperature.

12. What are the differences between the transport of materials in xylem and phloem?

See the xylem vs phloem table: xylem carries water and minerals upward only, by physical forces (root pressure, transpiration pull) without using energy; phloem carries food upward and downward through sieve tubes, using energy from ATP.

13. Compare the functioning of alveoli in the lungs and nephrons in the kidneys with respect to their structure and functioning.

See the alveoli vs nephrons table. In short: alveoli are thin-walled air sacs surrounded by capillaries that exchange O_2 and CO_2 by diffusion; nephrons are tubules with a glomerulus in Bowman's capsule that filter nitrogenous waste and selectively re-absorb useful substances. Both have thin walls, a large surface area and a rich blood supply.

Diagrams from the NCERT Science textbook for Class 10 (reprint 2026-27). Board questions from CBSE Class 10 Science papers 2023 to 2026.